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Menyanthes Quick Start Guide

KWR Watercycle Research Institute

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1 Main functions

1.1 Introduction

The program *Menyanthes* combines a variety of functions for managing, editing, visualizing, analyzing and modeling hydrogeologic time series. Within *Menyanthes,* time series can be modeled using both the ARMA and PIRFICT methods. The PIRFICT method is a new method of time series analysis that has practical advantages and facilitates physical interpretation and implementation of knowledge on physical behavior. Geohydrologic system analysis and interpretation of model results are supported by the spatial visualization and analysis tools of *Menyanthes*. Next to that, Menyanthes also has a well-developed suit of functions and tools for collecting, processing and managing data. These tools and functions are described in more detail in the following paragraphs.

1.2 Overview of available systems & services

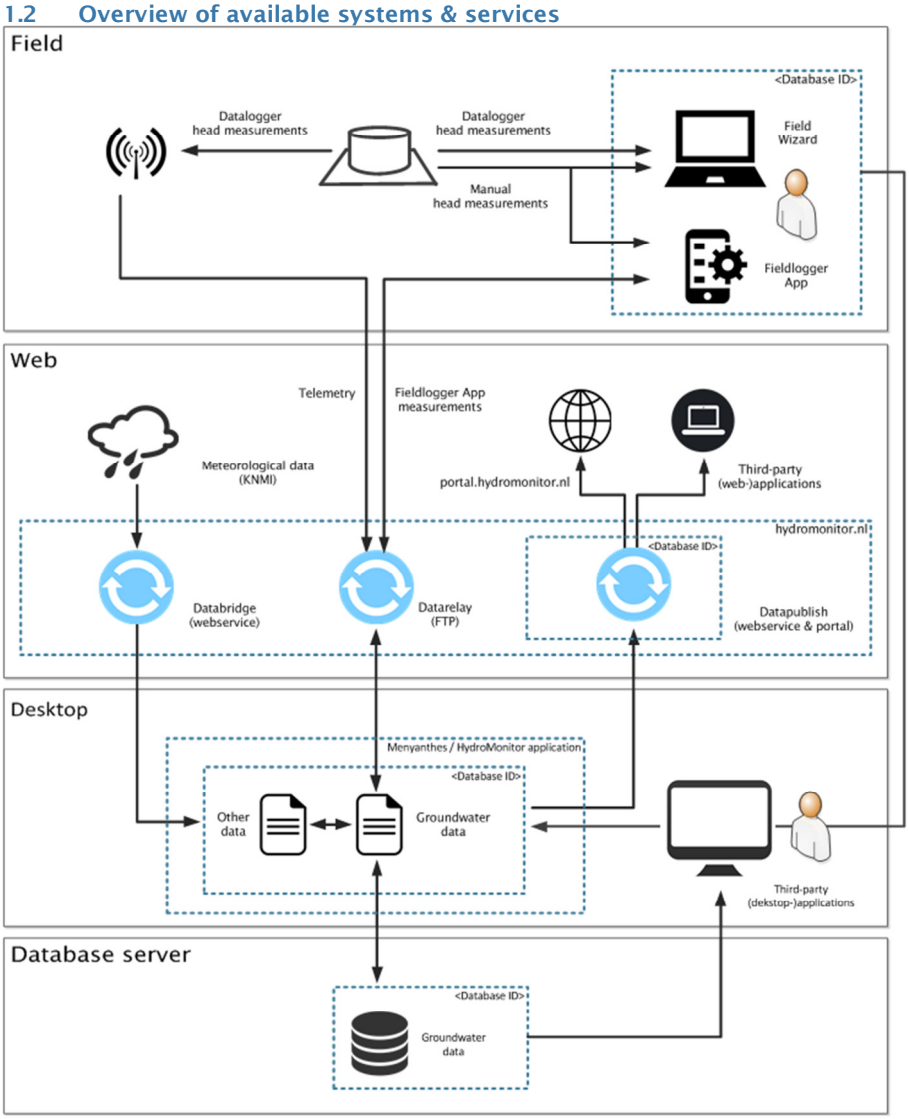


Figure 1: Overview of KWR's HydroMonitor systems & services and their interrelationships.

The Menyanthes desktop application is part of KWR's HydroMonitor systems & services. For field data collection, the so called Field Wizard and Field Logger App are available. For online data transfer and publishing, several web services are available. The resulting data may be stored in a local database file, or optionally on a remote database server. An overview of the available systems & services and their interrelationships is given in Figure 1.

1.3 Import and export

The Menyanthes desktop application supports data of several types, including groundwater head measurements and all data that are relevant for gaining insight in the functioning and properties of the geohydrologic system from which the measurements originate. These are:

- Measurement data:
  - Groundwater heads and levels
  - Surface water levels
  - Groundwater abstraction and/or infiltration rates
  - Meteorologic measurements

- Measurement locations and -instruments:
  - Observation wells and surface water level gauges
  - Data loggers and (pressure) sensors
  - Vegetation at the measurement location
  - A photo and/or situation sketch of the location

- Subsurface data:
  - Soil profile descriptions
  - Geohydrologic cross sections

- Surface data:
  - Aerial photo's
  - Topographic data
  - Digital Elevation Models

Menyanthes of course also allows you to export data and results for use outside of the program. This includes:

- Raw data
- Graphs, maps and other figures
- Model results
- Reports

1.4 Data collection and quality control (QC)

Next its time series analysis functions, Menyanthes also has a well-developed suit of functions for collecting, processing and managing data. These functions are part of the so-called HydroData module of Menyanthes, that is seamlessly integrated with the other functions. Functions specific to this task are :

- Data Collection:
  - Field data collection (via the so-called Field Wizard)
  - Import and processing of data logger files

- Data Quality Control (QC):
    - Automated checking (a.o. data integrity (QC1), Import and QC Wizard)
    - Visual checking (sensor data (QC2), consistency (QC3) and plausibility (QC4))

- Data Storage and Management:
    - Original data, corrections and comments
    - History of measurement locations and instruments

1.5 Visualization, analysis and modeling

Menyanthes functions whose importance cannot be easily overestimated are the presentation and visualization of series of measurements and spatial data. Presentation and visualization in general is done in:

- Graphs and tables
- Maps and cross sections

These functions support the other functionality where needed, and is not discussed in further detail here. Next to storing and presenting the raw data, Menyanthes also offers possibilities to perform simple to advanced operations on the data. These include:

- Processing operations for data collection purposes (e.g. air pressure compensation)
- Groundwater statistics (e.g. MxGL statistics, frequency of exceedence graphs, regime curves)
- Time series analysis and modelling

1.6 Help and support

KWR of course strives to let Menyanthes function flawlessly in any circumstance. If in spite of this a problem or error occurs, the following options are available under the 'Help' menu:

- Open the so-called error log file for showing error messages that may have occurred
- Request support by e-mail

2 User interface

2.1 Main user interface and quadrants

The main user interface opens automatically when Menyanthes is started. Initially, most of the functions and buttons are disabled. These functions become active when, for example, a database file (*.men) is loaded via File and Open (see Figure 2).

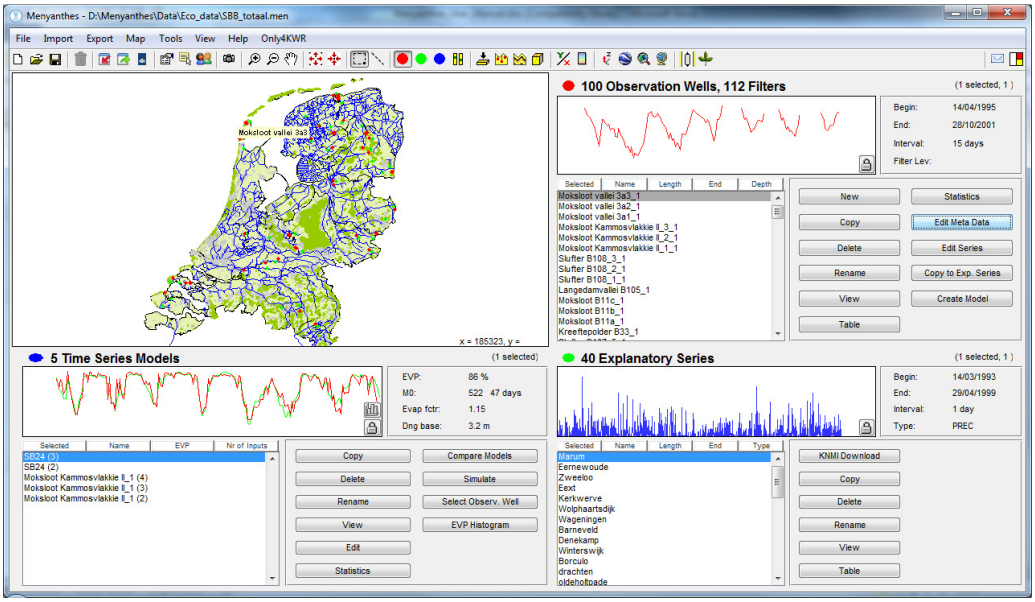


Figure 2: Main user interface of Menyanthes

Apart from the menu and toolbar, the main user interface consists of four quadrants:

1) Upper left - 'Map' showing a.o. the measurement locations
2) Upper left - 'Observation Wells' quadrant
3) Upper left - 'Explanatory Series' quadrant
4) Upper left - 'Time Series Models' quadrant

The default view of the main user interface is referred to as 'quadrant view' because of the four sections underneath the menu and toolbar. The contents of the four quadrants is discussed in paragraph 2.3.

2.2 The menu and toolbar

The menu bar and toolbar largely contain the same functionalities. For functions dealing with file management, select **File** or use the following buttons:

New – open a new database
Open – find and open an existing database
Save – save the current database

For functions related to the map, select **Map** or use the following buttons:

Zoom in/out – zoom in and zoom out with the left and right mouse button or by selecting an area (click and drag a rectangle).

Pan – Click and drag the map or graph.

Full zoom and zoom selected – zoom out to all observation wells or zoom to the selected wells only.

Select Rectangle – select locations with the mouse by dragging a rectangle around them

Select Transect – select locations along a line transect.

Select Observation wells, **Explanatory Series** or **Models** – Select the data type that 'Select Rectangle' and 'Select Transect' work on.

Top view – Show the locations in top view.

2D Cross-section – Show the locations in a 2D cross section view of the subsoil.

3D Cross-section – Show the locations in a 3D view of the subsoil.

Digital Elevation Model – Show digital elevation model of the area to which the map is zoomed.

Insert XY Labels – Insert axis labels or remove them.

Insert Color Bar – Insert color bar or remove it.

Virtual Earth – Insert background image from virtual earth or remove it.

Add/edit Topography – Insert topographic background and edit it.

Edit/Export Graph – Select an individual graph or a map and change the layout in another window and/or export the selected graph or map.

Additional functions can be accessed through the following buttons:

Spatial Analysis – Start the spatial analysis tool, which runs in a separate window.

Edit metadata – Open the meta data module to add or edit the elevations of the selected observation well or specify the serial number of the data logger.

Show boreholes – Plot borehole data (import supported for TerraIndex (*.xml) and Dino (*.xml, *.zip) files).

Edit Vegetation Tolerance – Start the vegetation reference module, which opens in a separate window.

The buttons at the right side of the toolbar are:

Contact support – Start your e-mail software for sending a support request to Menyanthes support.

Logger Data View – Switch to the Logger Data View instead of the Quadrant View, allowing you to manage and correct logger data,.

Quadrant View – Switch back to the Quadrant View of Menyanthes

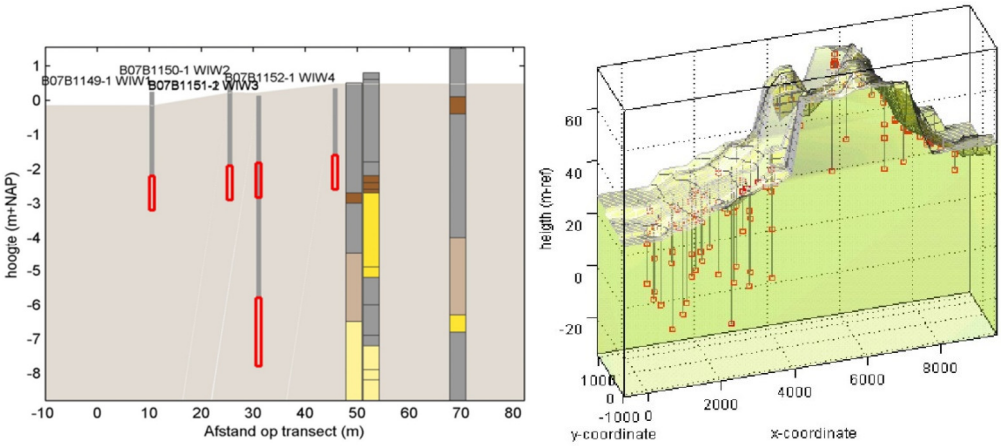


Figure 3: Cross sections of the subsoil in 2D and 3D

2.3 Quadrant list boxes and buttons

The Observation Well, Explanatory Series and Time Series Models quadrants have a similar layout and a number of buttons and options in common:

- A **list box**, allowing the user to scroll through the available series or models. When one or more series are selected, their position is shown on the map. Above the list box there are several buttons to sort the series or models on different properties.
- A **graph** showing the selected time series and/or model simulation.
- The **Lock time-axis** button next to the graph, for fixing the x (time) axis limits to the first and last observation of all series.
- A **text frame** with the main properties of the selected series or model.
- A **button panel** with the following common functions:
    - **Import** – import data of different formats into *Menyanthes*
    - **Export** – export data, statistics and results
    - **Copy** – copy the selected series or models
    - **Delete** – delete the selected series or models
    - **Rename** – rename the selected series or models
    - **View** – plot the selected series or models in a new window

The Observation Well quadrant contains buttons and functions concerning groundwater head series, including:

- A **Table** button to view the selected series in a table.
- An **Edit** module to adjust groundwater series and/or delete errors.
- An **Edit Meta Data** module to view and edit the technical data of a well.
- A **Statistics** module to calculate various groundwater level statistics.
- A **Create Model** module, to start the construction of a time series model.

The Explanatory Series quadrant contains buttons and functions for all other series. Most functionality has already been listed above, except for:

- A **KNMI Download** button for downloading meteodata from the Dutch meteorological institute KNMI.

The Time Series Models quadrant contains the following buttons and functions:

- The **Show residuals** button, above the 'lock time-axis' button next to the graph, to view the residuals instead of the simulation.
- A **View** button to open a new window with a graph of the observations, the predictions, and optionally either the residuals or the separate contributions of all different stresses.
- A **Statistics** module with the calibration statistics of the selected model.
- A **Compare Models** module to compare various characteristics of the selected models.
- A **Simulate** module to simulate groundwater level series using the selected model.
- A **Select GWL** button to select the corresponding groundwater level series in the 'Observation Wells' section.
- A **EVP Histogram** button, to produce an overview of the explained variances of the models in the database.
- A **Multi-plots** button to plot the simulations of different models in one figure.

3 Getting started

3.1 Database creation

Menyanthes stores all data on groundwater levels, explanatory series and time series models in one database (a local database file with *.men extension, or a database on a remote database server). During the installation process the subdirectory 'Data' is created in the installation directory in which you can save your database files. For creating a new database, select **File** or use the **New** button ( ) to create a new database.

Next, new locations can be added to the database using the **New** button in the Observation Wells quadrant. This button gives access to the so-called meta data interface, which is an interactive module for entering and editing construction data of a well (see *Figure 4*). In the **Preferences** section in the menu on the right, the preferences can be changed; you can choose between meters or centimeters as unit, and (national) reference level, surface level or top of the well as reference.

On the left in the metadata module, a picture of the well that is selected in the **Observation Well** list is shown. In the edit boxes at the right of the well, the values for surface level, top of observation well, top and bottom of the filter screen and the length of the sediment sump can be entered or edited. Under the heading **Coordinates** the coordinates of the selected well can be edited.

Via **Save** the new values are saved. The values are automatically recalculated and saved in meters with respect to the reference level. The old values are also saved, the dates of all entries and changes are shown in the list box under the heading **Date**. By selecting a certain date, the corresponding metadata becomes visible. Old metadata, however, can no longer be edited.

When a value is entered or changed, there is an instant check on the **data validity**. For example, the surface level and the top of the well cannot be lower than the (bottom) level of the filter screen, and the top of the filter screen must be higher than the bottom.

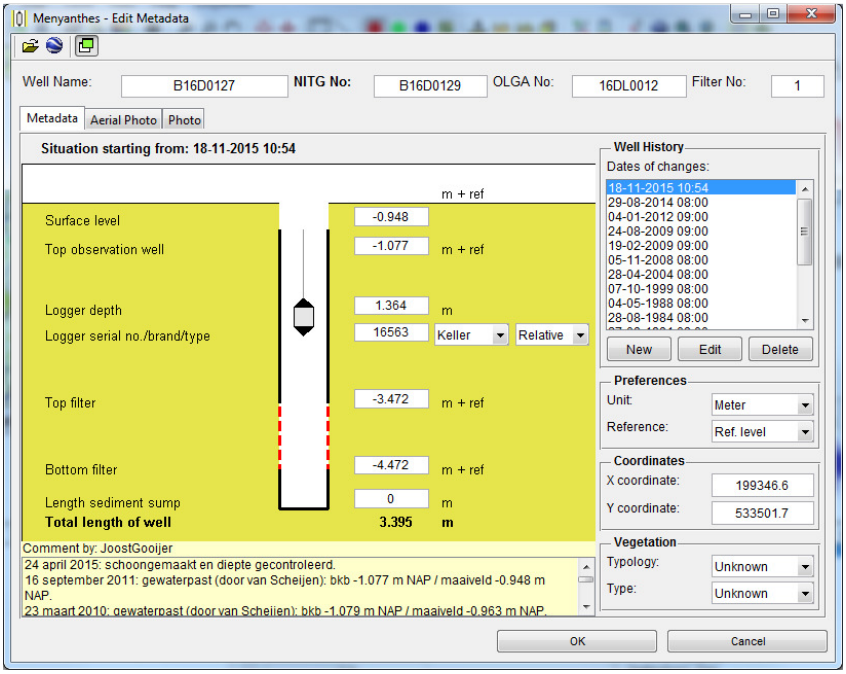


Figure 4 Module for editing the construction or meta data of a well

3.2 Import formats

In spite of initiatives for developing open standards for data models and formats, specialized software like Menyanthes cannot do without its own data exchange format. The main format that Menyanthes uses is called the HydroMonitor format with properties:

- Internal definition of format name and version
- Interne definition of format variation (file contents, object type, location ID label or field)
- Interne definition of data columns and fields (data type, unit, reference level)
- More details and elaborate definitions are documented and available via a link in the data file itself
- Available in Excel, CSV and HDF5 / Matlab form
- Supported for import and export

More details about this format are contained in a document that is available via the following link:

http://www.hydromonitor.nl/downloads/hydromonitor_data_exchange_format.pdf

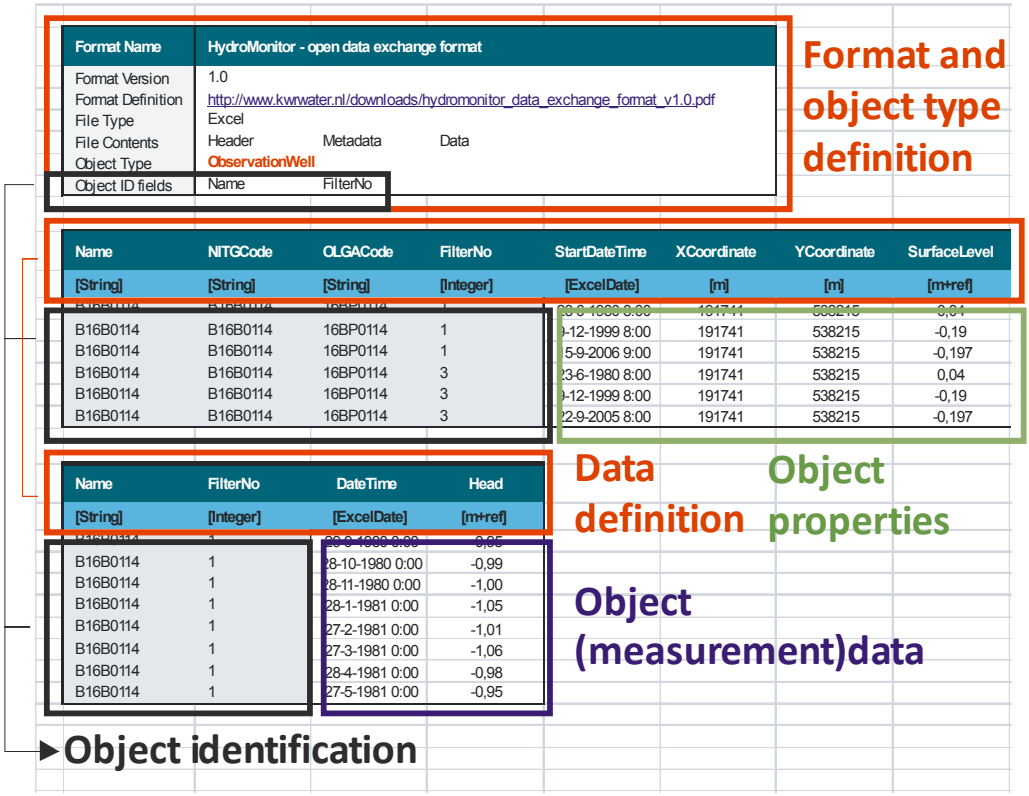


Figure 5: Example of the HydroMonitor format, explaining its different elements and properties.

Next to the HydroMonitor format, Menyanthes supports a range of other widely used data formats. An overview and examples of these formats may be downloaded via **Help** > **Download** > **File Format Examples.**

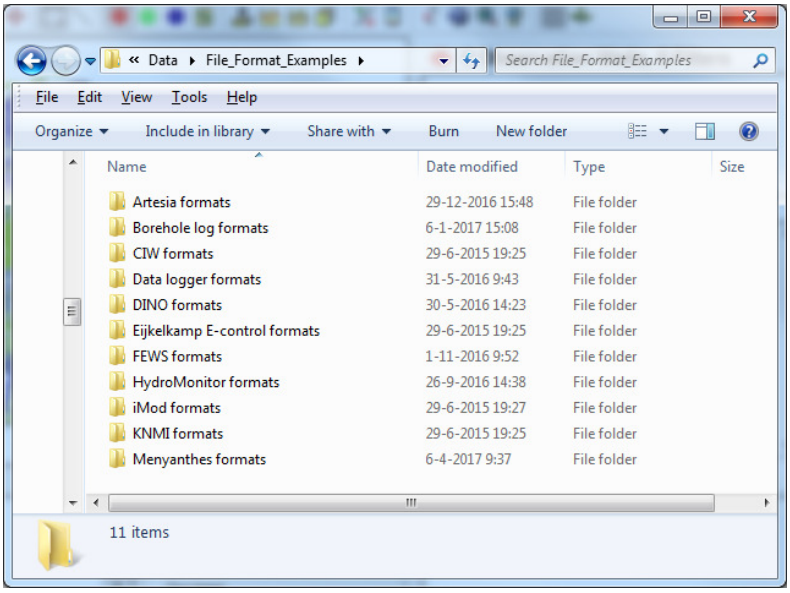


Figure 6: Downloaded and unzipped directory containing an overview and examples of the formats that Menyanthes supports.

3.3 Location ID's and labels

As is visible in *Figure 4* and *Figure 5*, Menyanthes supports the use of multiple labels for the same location or observation well. This may be valuable, as the same well may be registered in multiple databases, and may be labeled and be known in different ways by different organizations. To be specific, the 'Name' field is used in general for the local label of a well, as attributed by and known to the local organization or well owner, whereas both the 'NITGCode' and 'OLGACode' fields are labels that are attributed on a (Dutch) national level and databases.

For identification purposes, however, a requirement is that all Observation wells are labeled uniquely and completely by at least one of these labels. In Menyanthes, the so-called 'Primary Label' that is used for identifying all wells may be set and selected in the options interface. By selecting e.g. the 'Name' field as primary location label, the 'Name' will be used as standard label in user interfaces and for purposes such as graphs and time series models. The other labels, may be made visible also in the listbox and/or map, by selecting that field in the listbox with a right button click (*Figure 7*).

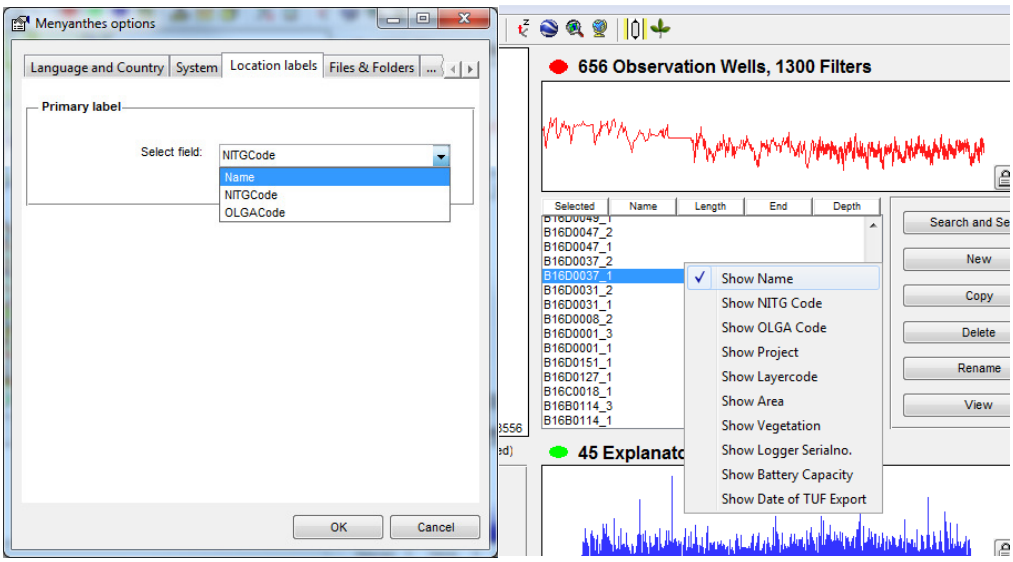
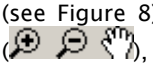


Figure 7: Selection of the location label as 'Primary Label' in the options interface (left) and for display in the listbox (right)

3.4 Time series editing

When the **Edit** button in the Groundwater section of the main window is pressed, a new window opens in which you can interactively edit and delete groundwater level observations (see Figure 8). The buttons in the toolbar give the options to zoom in, zoom out or pan (), or to make a selection of the observations ( ). The values of the selected observations are shown in the table at the right. **Invert Selection** gives the option to invert the selection. Use **Delete** to delete the selected observations. Using the '+/-'button to add or submit an arbitrary value from the selected observations (this is for example useful for '1 meter errors' and other typing errors in the data). The '*' button may be used to multiply selected values. The '=' button may be used to assign a certain value to the selected observations. In the **Period** edit box you can enter two dates to select data from a certain period, while level gives the option to select all data within a range. Click **Save** to save the

adjusted values in the *Menyanthes* database. The buttons may be used to undo or redo changes.

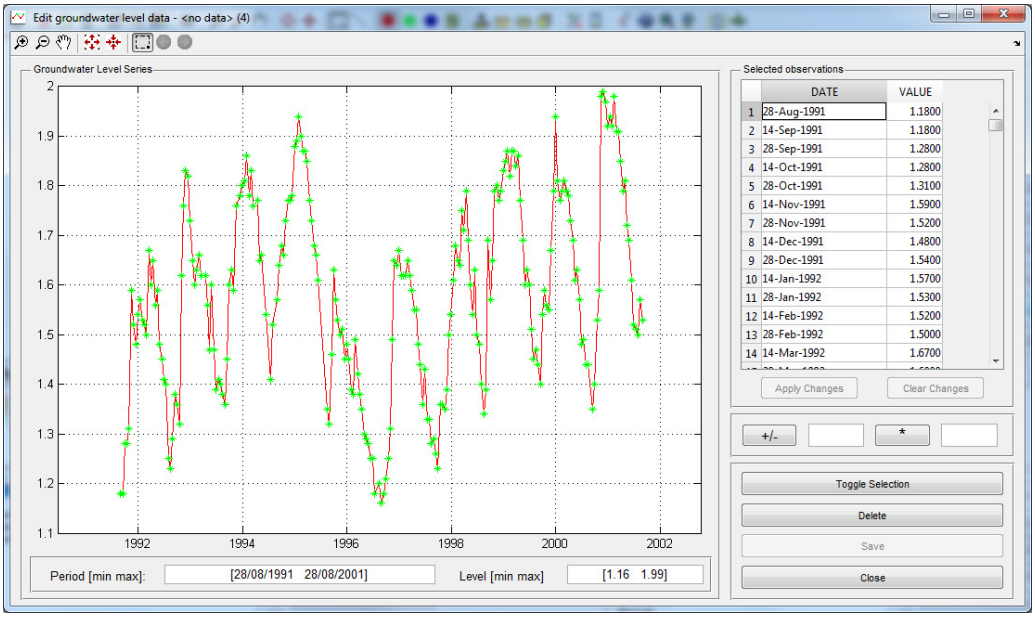


Figure 8: User interface for editing time series data

4 Groundwater system analysis

4.1 Introduction

As functionality, Menyanthes combines tools for both system analysis and time series analysis The fact that Menyanthes contains functions for both is not a coincidence. Where one may define 'system analysis' as gaining qualitative insight in the main structure, functioning and properties of a system, the term 'system identification' generally refers to gaining quantitative insight on these matters and is used more or less as a synonym for time series analysis (see *Figure 9*). In practice, a project involving time series analysis with Menyanthes usually starts with an exploration of the system under consideration, in combination with an explorative look at the measurement data themselves.

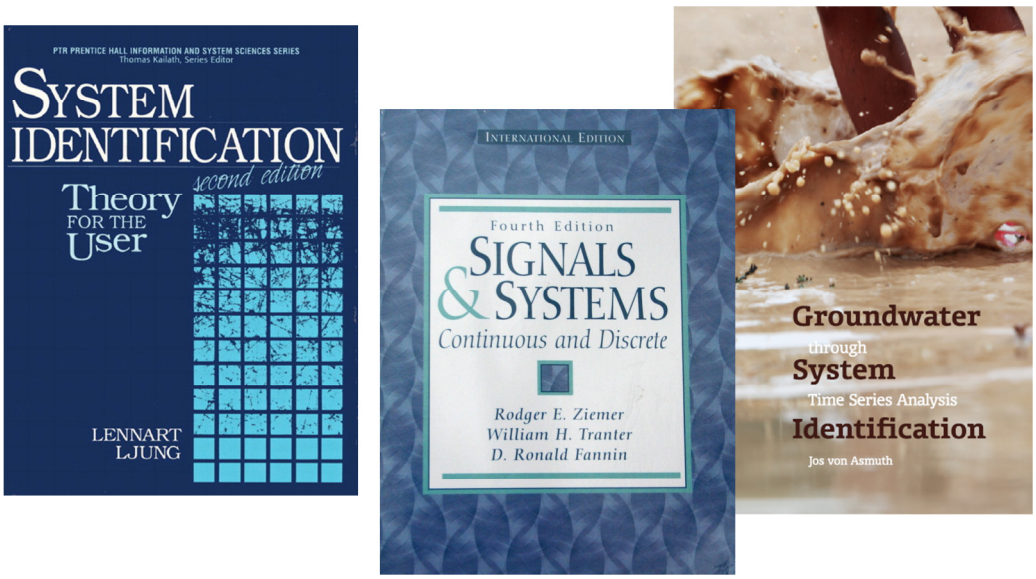


Figure 9: Examples of technical-scientific literature on system identification and time series analysis (Ziemer e.a., 1998; Ljung, 1999; Von Asmuth, 2012)

4.2 Visualization and analysis

In paragraph 1.3, an overview was given of the data types that Menyanthes support, many of which are aimed at gaining insight in the structure, functioning and properties of geohydrologic systems. Menyanthes provides a variety of tools to present and visualize the this data in maps (surface and subsurface data) and 2D or 3D cross sections (subsurface data). Visualization examples are presented in *Figure 10*.

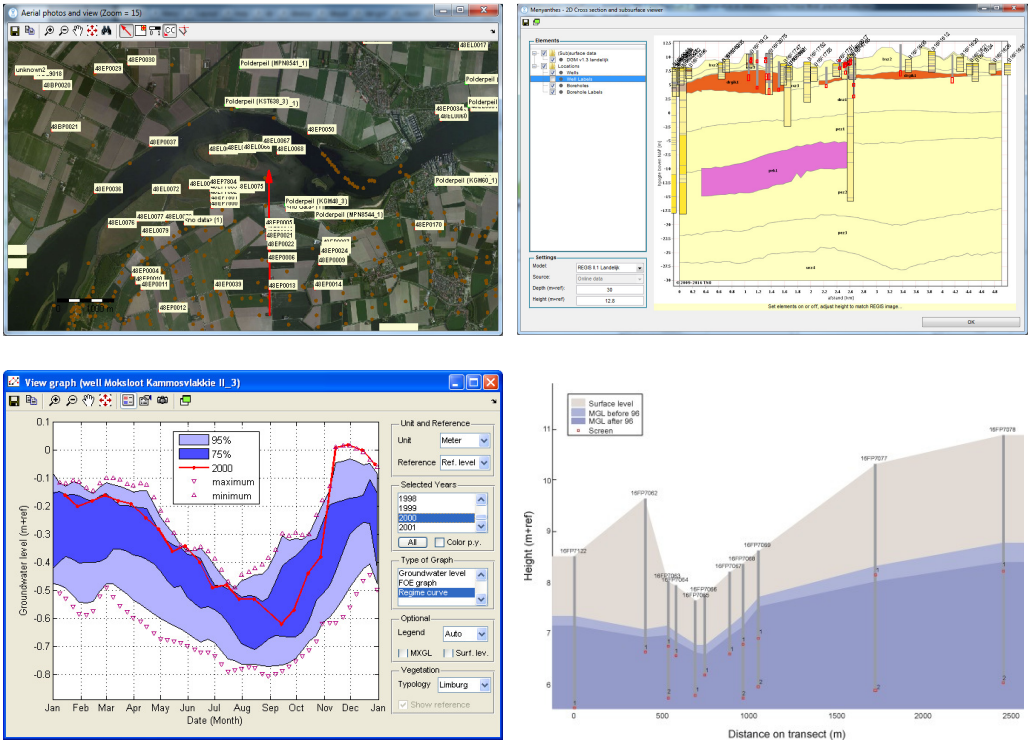


Figure 10: Examples of visualization and analysis tools in Menyanthes. (a) Aerial photo, topography, measurement locations and transect, (b) 2D cross-section with soil profiles and subsurface model, (c) groundwater head regime curve (b) 2D cross-section showing the course of the average groundwater head.

4.3 Conceptual model or schematization

Using the insight gained from visualizing and analyzing the available surface, subsurface and monitoring data, a conceptual model or schematization of the system under consideration may be derived (see e.g. Figure 11). Drawing up a conceptual model is a good way of summarizing the insights into the structure and processes that occur in a system and interact with it. A conceptual model provides a natural step towards deciding on the time series model approach and series to use.

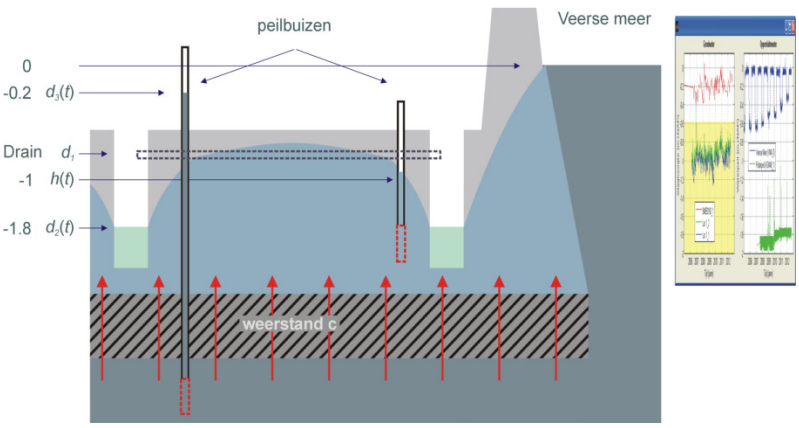


Figure 11: Example of a conceptual model or schematization of a polder system near the lake Veerse Meer in the southwest part of the Netherlands, derived from the available surface, subsurface and monitoring data (von Asmuth (Von Asmuth en Leunk, 2012)

5 Time series modeling

5.1 Introduction

The development of Menyanthes initially started within the scope of a research project, whose primary aim was the integration of data and physically-based methods for modeling time series of groundwater heads. As such, time series analysis still forms the heart of Menyanthes and the set-up and coloring of the main user interface follows the schematic representation of a system as used in system and time series theory.

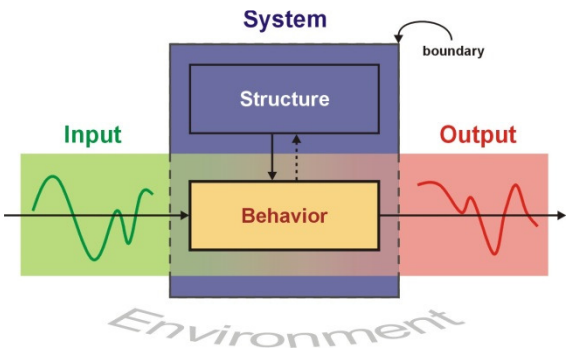


Figure 12: Schematic representation of an open physical system (Von Asmuth, 2012). The set-up and coloring of the main user interface of Menyanthes follows this schematization, which adds to the intuitiveness of the user interface.

Within Menyanthes, time series can be modeled using both the ARMA and PIRFICT methods. The PIRFICT method has practical advantages over the more traditional ARMA method, and facilitates physical interpretation and implementation of knowledge on physical behavior. The PIRFICT method allows for the use of a standard response function for a specific type of input, and model identification consequently reduces to the problem of identifying which inputs or stresses influence the dynamics of a system (Von Asmuth et al., 2008). In case of groundwater heads, or any environmental variable for that matter, stresses tend to have a spatial effect and influence all locations within a certain region. Assessment of the region of influence of a certain stress (for instance groundwater extraction) is in fact one of the possible applications of time series models. Using the PIRFICT method, stress identification therefore generally applies to all observation wells in an area, making it a batch process.

The methods related to the PIRFICT approach are described in several papers. It was shown that the PIRFICT method allows for the incorporation of physical behavior, reduces the number of parameters, simplifies model identification and avoids problems related to the data frequency in time series models (Von Asmuth e.a., 2002; Von Asmuth en Bierkens, 2005). The PIRFICT method was extended and tested to include complex situations with multiple stresses of different types (Von Asmuth e.a., 2008). The use of time series model results and/or impulse responses in distributed models was addressed in (Von Asmuth en Maas, 2001; Bakker e.a., 2007; Bakker e.a., 2008). Finally, in (Von Asmuth e.a., 2012) a synthesis is given of the methods, the program *Menyanthes* is presented and an example

application is given where data from multiple locations are analyzed, illustrating the use of spatial patterns and physical insight to interpret results.

5.2 Creating a time series model

The procedure for modeling groundwater level series starts in the 'Observation Wells' section of the main window. First, select the series to be modeled in the list box, and then press the **Create Model** button. If coordinates are present in the data set, Menyanthes will automatically select the nearest precipitation and evaporation stations as explanatory series, for every series. If not, the program will use the series that are selected in the explanatory series list box. After this the 'Identify Time Series Model' user interface appears (Figure 13).

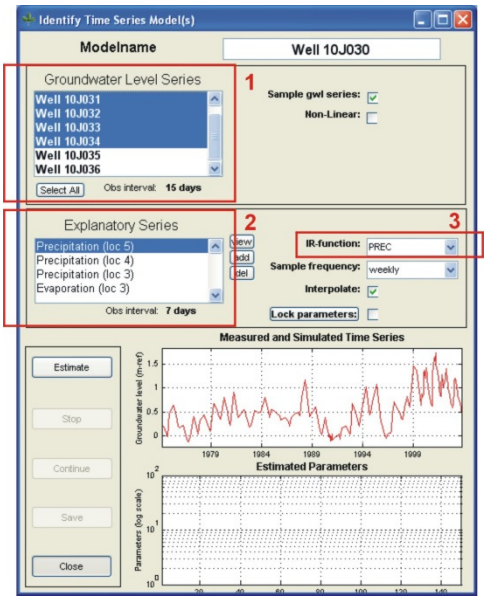


Figure 13: *User interface for 'identifying' multiple time series models simultaneously.*

This user interface supports the process of simultaneous model identification for multiple time series. In the box marked with a one, there is a listbox with locations or groundwater level series. All settings made in the user interface apply to the series selected in this listbox. In the box marked with a two, there is a listbox with stresses or explanatory series. *Menyanthes* automatically selects the nearest meteorological stations, as precipitation and evaporation always influence the groundwater level. The explanatory series shown in listbox two are not necessarily used for *all* locations in listbox one, but in contrast, it is a list of explanatory series used for *any* of them. Properties of explanatory series that are set, are set for all locations selected in listbox one for which they are used. An example is the type of impulse response function used for a certain stress (see Von Asmuth et al., 2008), which can be selected in the drop-down menu that is marked with a three. When appropriate settings, explanatory series and response functions are chosen, the model parameters can be optimized by pressing the 'estimate' button, after which the models are optimized one-by-one. Details on the optimization procedure and noise model, which are also based on a continuous-time formulation, are presented in (Von Asmuth and Bierkens, 2005).

5.3 Influence and impact analysis

Standard time series models are linear, and in case a system behaves significantly non-linear, alternative time series models specifically dealing with that should be used. Linearity, however, also has a number of advantages. A consequence of linearity is that time series models can decompose time series into partial series showing the effects of individual inputs (*Figure 14*). This property enables intervention analysis as described by (Hipel e.a., 1975; Hipel en McLeod, 1994). Time series are decomposed in an 'explained' and an 'unexplained' part (even for the single input case), allowing for trend detection or the quantification of anthropogenic influences when there is only one natural factor such as precipitation surplus (e.g. (Rolf, 1989; Gehrels, 1999; Yihdego en Webb, 2011). In the case of multiple inputs, the primary interest is usually the impact of one of these inputs, such as pumping (e.g., (Von Asmuth e.a., 2008; Harp en Vesselinov, 2011) or hydrologic measures. More in general, time series analysis can reveal much of the functioning of a system and of the interactions between its parts (e.g., (Lee en Lee, 2000; Fabbri e.a., 2011). An archetypical application of time series models lies in the area of 'control engineering', where the problem is how to manipulate the inputs into a system to obtain the desired output. For hydrogeologic purposes, this means that time series models can be used to optimize (or identify problems with) the water management in an area (e.g., (Bidwell e.a., 1991; Lee e.a., 2005). An example of a series that was decomposed into several partial series showing the influence of different inputs or explanatory series is shown in Figure 14 .

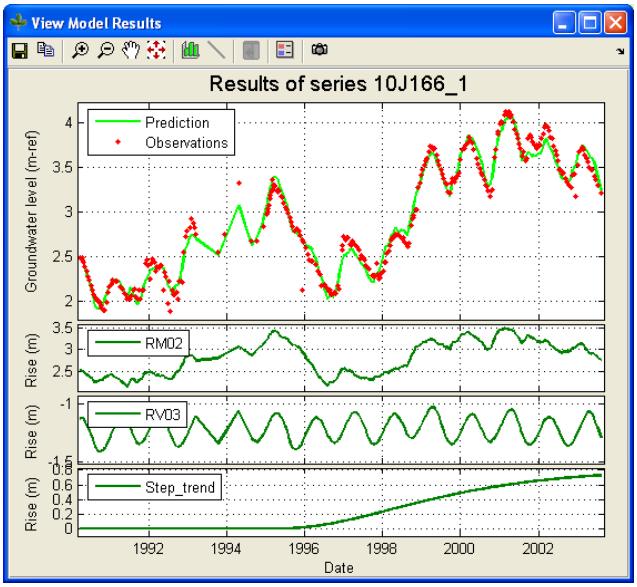


Figure 14: Groundwater level series decomposed into partial series showing the influence of individual inputs (Von Asmuth e.a., 2012)

5.4 Interpretation and spatial analysis

As mentioned, PIRFICT models may be fitted to multiple time series in batch. Spatial patterns that emerge in the results provide useful, additional, and independent information, which adds another dimension to time series analysis. Their interpretation is supported by the spatial visualization and analysis tools of Menyanthes. In *Figure 15*, several visualization examples showing spatial patterns in results from multiple series models in 2D cross-sections are given. For details about their interpretation, see *(Von Asmuth e.a., 2012).*

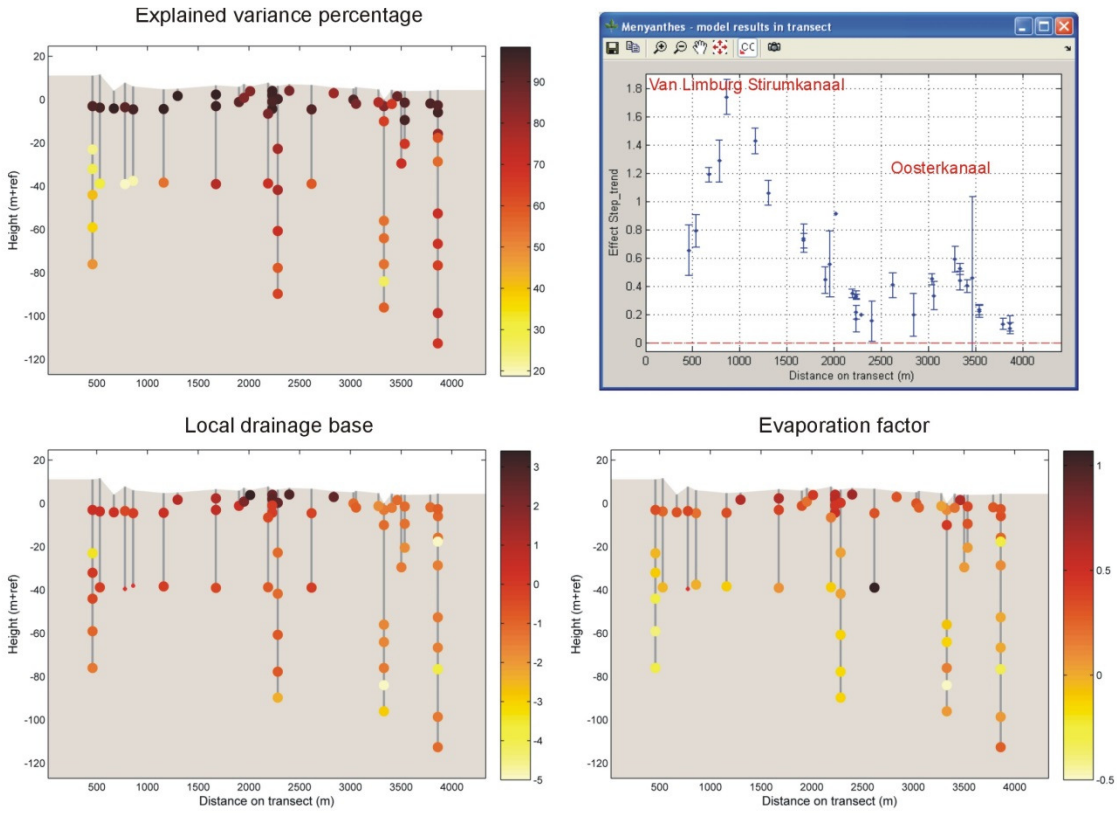


Figure 15: Model results along a transect, showing a) the explained variance percentages, b) the estimated effect, c) the local drainage base, d) the estimated evaporation factor, all as colored dots at the locations of the observation well screens (Von Asmuth e.a., 2012) .

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Appendix I Technical specifications and requirements

Introduction

Menyanthes is developed in a MATLAB environment, with which it is compiled into a standard 32-bit Windows application. The so called MATLAB Runtime, or libraries needed to run Menyanthes, are packaged together with Menyanthes into a single installer file (executable). Both, however, are also available separately. Menyanthes can be installed stand-alone, as well as distributed over a network.

Licensing

There are two license options. Menyanthes operates either with a network domain license (access on all computers within a Windows domain) or with a 'Single Computer License' for individual laptops, tablets or PC's. In case Menyanthes is used on a virtualized computer, it may only be used by a single user or person. For more information please see chapter 2 'installation procedure'.

System requirements

The basic system requirements of Menyanthes are identical to those of MATLAB (r2014a) and are copied beneath from the website of the Mathworks (the developers of MATLAB), for sake of convenience (*). The only adjustment made is an increase in the RAM amount, recommended for processing large data amounts:

Operating System	Windows XP SP3, x64 Edition SP2, Windows Vista SP2, Windows 7 SP1, Windows 8 or 8.1, Windows Server 2003 SP2(R2), Windows Server 2008 SP1(R2) or Windows Server 2012 SP2.
Processor	Any Intel or AMD x86 processor supporting SSE2 instruction set*
Disk Space	1 GB (minimum, Menyanthes only)
RAM	2048 MB (Minimum), 4096 MB (recommended when processing large data amounts)
Graphics	No specific graphics card is required. Hardware accelerated graphics card supporting OpenGL 3.3 with 1GB GPU memory recommended.

* More information is available at:

http://www.mathworks.com/matlabcentral/answers/93455-what-is-the-sse2-instruction-set-how-can-i-check-to-see-if-my-processor-supports-it

Virtualized operating systems

The Mathworks do not officially test and support the functioning of MATLAB on operating systems virtualized e.g. with Citrix or VMware. As KWR and Menyanthes depend on technical support from the Mathworks for any technical issues related to MATLAB or the MATLAB Runtime, also the correct functioning of Menyanthes under virtualized operating systems cannot be guaranteed. However, Menyanthes is used internationally by a wide range of users (e.g. engineering offices, governmental institutes, universities) on a variety of platforms without any problems, including virtualized systems.

Network installation and distribution

KWR does not test the installation of every Menyanthes release under every operating system included in the system requirements, nor does KWR itself have direct access to all these operating systems. The performance and distribution of Menyanthes on ICT networks are the responsibility of you as a Menyanthes user, or your ICT organization. On line support from KWR and troubleshooting of installation issues is available using the program Teamviewer, that is distributed with Menyanthes. In general, support is available only for users having an up-to-date Service Agreement. For technical issues related to MATLAB or the MATLAB Runtime libraries, technical support from the Mathworks may be requested. KWR, however, will be your primary contact in such cases.

Third party products and components

For specific tasks, Menyanthes relies on third party products and software components, which may have their own system requirements and for whose correct functioning KWR is not responsible. These include:

- Microsoft Excel 2000 or higher for importing and exporting data (*.xls, *xlsx)
- Microsoft Word 2000 or higher for generating Word documents (*.doc)
- The program Teamviewer for long distance support from KWR on Menyanthes
- The Dracal USB Precision Barometer and accompanying driver
- The Navilock NL-601US USB GPS receiver and accompanying driver
- A selection of water level and air pressure loggers (e.g.. Diver, Keller) and accompanying software

Appendix II Installation and activation

Available installers

The standard installer for Menyanthes installs the entire application, as well as the MATLAB Runtime. If you have Menyanthes installed already, and you want to update to the latest version, you can also use the update package.

Full installation

This section describes the installation procedure when Menyanthes is installed as a standalone, single user application on an individual laptop or PC. Menyanthes is installed with the executable installation package. Upon the first usage, the user is prompted with a license registration procedure that has to be completed. When an older Menyanthes version (or the corresponding MATLAB version) is already installed, a so-called update or no-runtime version can also be used that does not require full installation.

To install Menyanthes please follow the following step-by-step procedure:

- Download the installation wizard through the link send to you by e-mail.
- In Windows Explorer double click the downloaded setup file.
- The setup procedure will start automatically, and will guide you through the installation process.
- After the installation is complete, start Menyanthes by clicking the desktop icon or the Menyanthes icon in your start menu.

Update or no-runtime version

If Menyanthes is already installed on your laptop or PC, you can avoid deinstalling the existing version and reinstalling the new version, by using the so-called 'no runtime' version. This version only contains Menyanthes itself and not the full MATLAB runtime libraries it depends on. Consequently, the no-runtime version has a significantly smaller file size. This version is available as a ZIP-file with a name like:

Menyanthes v.3.x.x.x (no runtime).zip

If you unzip this file onto a PC where Menyanthes is already installed you can start and use the new 'Menyanthes.exe' without further action (if desired using a shortcut to it on the desktop). You are free to choose the location to place the no runtime version in (even a USB-stick suffices), with the exception of the directory where the existing version is installed.

License activation

Standalone license

If you run Menyanthes for the first time you have to register your license. After start up, a message similar to the one below will appear:

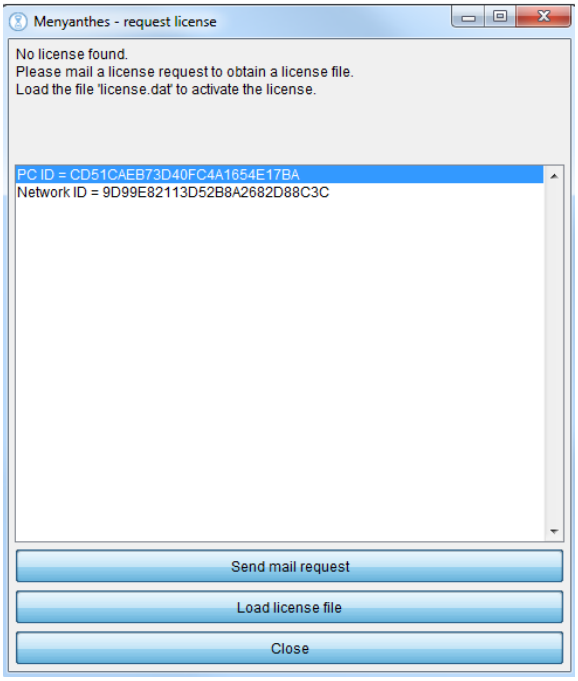

Figure 1: Menyanthes license check

For a stand-alone / PC-license, please send the PC-ID code to KWR by clicking the 'Send mail request' button. After submission, you will receive a 'license.dat' file containing the actual license by e-mail (this license file will be send to you by the Menyanthes support staff). If you are not connected to the internet or have trouble sending the PC ID with the program, you can also send the ID to *menyanthes_support@kwrwater.nl*. With the license file you can register your version of Menyanthes by saving the file on a local or network drive and clicking the 'Register' button, and get started.

Attention: If you run Menyanthes in order to obtain a code for a PC-license, make sure you do it on the computer where the program will actually be used.

Network license

The network license is linked to the Security Identifier of the Windows NT Domain (i.e. the domain of which the users are a member, not the domain the computers are linked to). All recent versions of MS Windows (except XP Home Edition) offer this facility. After registration, Menyanthes can be used when the user has logged onto Windows, even without actual connection to the network (for example when using a laptop offline).

When installing Menyanthes on multiple PC's in a network using an installation script, the file 'License.dat' can be installed along with the other program files and be placed in the Installation directory (e.g. 'C:\Program Files (x86)\Menyanthes'). After doing so, Menyanthes will automatically register the license, so this does not have to be done manually on every individual PC.

Attention:

- Menyanthes needs to be installed locally on every machine, even though the license is linked to the network domain.

- Before you run Menyanthes in order to get an ID code for a network license, please be sure that you are logged in to the Windows Domain the actual users are also a member off (this is not the Domain to which the computer is linked).
- If different Menyanthes users belong to different Windows Domains within your organization, you will need different license files for each Domain.

Installation troubleshooting

'Application has failed to start because mclmcrrt8_3.dll was not found'
This message appears when the Menyanthes 'PATH' variable has not been set properly. You can verify this by checking whether the files mclmcrrt8_3.dll is present in the directory:

[installdir]\Menyanthes\MCR\v83\runtime\win32

This error may occur if the installation has been carried out from the Windows Explorer after it has been started using the 'runas' option to get administrator rights. In this case, the installation sets the path for the administrator, but not for the user. This can be fixed manually by adding the proper directory to the system path variable or by defining a user environment variable with the name 'PATH'. Under windows XP these variables can be accessed via the 'System Properties' on the tab 'Advanced' using the button 'Environment Variables'. You can get to the System Properties by clicking with the secondary mouse button on the icon 'My Computer' on the desktop or in the windows explorer and selecting properties, or via the Start Menu: Settings | Control Panel | System.

Write permissions, preferences and error messages
Menyanthes writes user preferences and other data to a file named 'prefs.mat'. You can find this file in:

[AppData dir]\[Roaming or Local]\Menyanthes

Menyanthes requires its users to have write permissions in this directory. Possible error messages of Menyanthes are written to the following file:

[AppData dir]\[Roaming or Local]\Menyanthes\errordump.txt

Menyanthes therefore also requires write permission in this directory.